

9. 原式左、右兩側同乘 $(x-3)(x+2)$ ：

$$4x-7=A(x+2)+B(x-3)$$

- (1) 令 $x=-2$ 代入：

$$4 \times (-2) - 7 = A \times (-2+2) + B \times (-2-3)$$

$$\Rightarrow B=3$$

- (2) 令 $x=3$ 代入：

$$4 \times 3 - 7 = A \times (3+2) + B \times (3-3)$$

$$\Rightarrow A=1$$

$$\text{故 } 2A+3B=2 \times 1+3 \times 3=11$$

$$10. \because (a-b)(a^2+ab+b^2)=a^3-b^3$$

$$\begin{aligned} \therefore (\sqrt[3]{3}-2)(\sqrt[3]{9}+2\sqrt[3]{3}+4) \\ &= (\sqrt[3]{3}-2) \left[(\sqrt[3]{3})^2 + \sqrt[3]{3} \times 2 + 2^2 \right] \\ &= (\sqrt[3]{3})^3 - 2^3 = 3 - 8 = -5 \end{aligned}$$

10 單元 5 直線與圓

(主題 1~13：直線方程式)

答案

1. $m_1 < m_3 < m_2$
2. 1
3. $2x-y-10=0$
4. $2x+3y+3=0$
5. 9
6. -49
7. $-\frac{7}{3}$
8. $2x-y+1=0$
9. 2
10. 36

$$1. m_1 = \frac{3-(-3)}{\frac{1}{2}-\frac{7}{2}} = \frac{6}{-3} = -2, m_2 = -\frac{1}{-3} = \frac{1}{3}$$

$\because y=3$ 是水平線 $\therefore$ 其斜率 $m_3=0$

$$\because -2 < 0 < \frac{1}{3} \therefore m_1 < m_3 < m_2$$

$$2. A、B \text{ 連線斜率} = \frac{2-6}{10-(-2)} = \frac{-4}{12} = -\frac{1}{3}$$

$$B、C \text{ 連線斜率} = \frac{(a+4)-2}{a-10} = \frac{a+2}{a-10}$$

$\because A、B、C$ 在同一直線上

$$\therefore \frac{-1}{3} = \frac{a+2}{a-10} \Rightarrow 3a+6 = -a+10$$

$$\Rightarrow 4a=4 \Rightarrow a=1$$

$$3. \text{由點斜式, } y-(-2)=2 \times (x-4)$$

$$\Rightarrow y+2=2x-8 \Rightarrow -2x+y+10=0$$

$$\stackrel{\times(-1)}{\Rightarrow} 2x-y-10=0$$

$$4. \text{由斜截式, } y = -\frac{2}{3}x + (-1)$$

$$\stackrel{\times 3}{\Rightarrow} 3y = -2x - 3 \Rightarrow 2x + 3y + 3 = 0$$

5. $\because$ 直線 L 的 x 截距為 3

$\therefore$ 直線 L 通過點 $(3,0)$

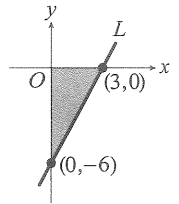
由點斜式

$$L: y-0=2 \times (x-3)$$

$$\Rightarrow 2x-y-6=0$$

x	3	0
y	0	-6

$$\text{因此所求三角形之面積} = \frac{1}{2} \times 3 \times 6 = 9$$



$$6. (1) \frac{x}{y} \left| \begin{array}{cc} 28 & 0 \\ 0 & 7 \end{array} \right. \quad x \text{ 截距 } a=28, y \text{ 截距 } b=7$$

$$(2) \text{直線 } L: x+4y=28 \text{ 的斜率 } m = -\frac{1}{4}$$

$$\text{故 } abm = 28 \times 7 \times \left(-\frac{1}{4}\right) = -49$$

7. (1) 直線 L 通過 $A、B$ 兩點

$$\text{則斜率} = \frac{2k-2}{1-(-k)} = \frac{2k-2}{k+1}$$

$$(2) \text{直線 } L_2: x+5y-5=0 \text{ 的斜率} = -\frac{1}{5}$$

$$\because L \perp L_2 \therefore \frac{2k-2}{k+1} \times \left(-\frac{1}{5}\right) = -1$$

$$\Rightarrow 5k+5=2k-2 \Rightarrow 3k=-7$$

$$\Rightarrow k = -\frac{7}{3}$$

8. 設直線 $L: 4x-2y+k=0$

$$\text{點 } (0,1) \text{ 代入 } L: 4 \times 0 - 2 \times 1 + k = 0 \Rightarrow k=2$$

$$\text{故直線 } L: 4x-2y+2=0 \stackrel{+2}{\Rightarrow} 2x-y+1=0$$

$$9. (1) \text{直線 } \overleftrightarrow{BC} \text{ 的斜率} = \frac{3-(-2)}{-7-5} = -\frac{5}{12}$$

$$(2) \overleftrightarrow{BC}: y-(-2) = -\frac{5}{12}(x-5)$$

$$\Rightarrow 5x+12y-1=0$$

- (3) 點 $A(3,1)$ 到 $\overleftrightarrow{BC}$ 的距離

$$= \frac{|5 \times 3 + 12 \times 1 - 1|}{\sqrt{5^2 + 12^2}} = \frac{26}{13} = 2$$

10. 設 $L_1: 3x+4y+k_1=0, L_2: 3x+4y+k_2=0$

$$\text{點 } (-29,23) \text{ 代入 } L_1: 3 \times (-29) + 4 \times 23 + k_1 = 0$$

$$\Rightarrow k_1 = -5$$

$$\text{點 } (31,23) \text{ 代入 } L_2: 3 \times 31 + 4 \times 23 + k_2 = 0$$

$$\Rightarrow k_2 = -185$$

因此兩平行線間的距離

$$= \frac{|k_1 - k_2|}{\sqrt{3^2 + 4^2}} = \frac{|-5 - (-185)|}{5} = \frac{180}{5} = 36$$

11 單元 5 直線與圓

(主題 14~22: 圓方程式、圓與直線關係)

答案

1. $(x-2)^2 + (y+2)^2 = 25$
2. 4
3. 6
4. $k > -5$
5. $\begin{cases} x = 3 + 5\cos\theta \\ y = -4 + 5\sin\theta \end{cases}, 0 \leq \theta < 2\pi$
6. 圓上
7. 4
8. $2\sqrt{11}$
9. $3x + y - 9 = 0$
10. $4x + 3y - 27 = 0$

1. 圓 C 的圓心是 $\overline{AB}$ 的中點

$$\frac{A+B}{2} = \frac{(5,2) + (-1,-6)}{2} = \frac{(4,-4)}{2} = (2,-2)$$

半徑是圓心到 A 點的距離

$$= \sqrt{(5-2)^2 + [2-(-2)]^2} = 5$$

$$\text{故圓 } C: (x-2)^2 + [y-(-2)]^2 = 5^2$$

$$\Rightarrow (x-2)^2 + (y+2)^2 = 25$$

2. 如圖，兩直線 L_1 與 L_2 的距離為 $b_1 - b_2$ 也是圓 C 的直徑

而圓 C 的半徑為 2，故 $b_1 - b_2 = 2 \times 2 = 4$

3. (1) 圓 $C_1: [x-(-6)]^2 + [y-(-2)]^2 = 2^2$
其圓心 $(-6, -2)$ ，半徑 $r_1 = 2$

$$(2) \text{ 圓 } C_2 \text{ 的圓心 } \left(-\frac{12}{2}, -\frac{-6}{2}\right) = (6, 3)$$

$$\text{半徑 } r_2 = \frac{1}{2} \sqrt{(-12)^2 + (-6)^2 - 4 \times 20} = 5$$

- (3) 二圓心的距離

$$d = \sqrt{[6-(-6)]^2 + [3-(-2)]^2} = 13$$

$$\text{故 } d - r_1 - r_2 = 13 - 2 - 5 = 6$$

4. 圓: $x^2 + y^2 + 2x + 4 + 4y - 4 - k = 0$

$$\Rightarrow x^2 + y^2 + 2x + 4y - k = 0$$

$$\text{則 } 2^2 + 4^2 - 4 \times (-k) > 0$$

$$\Rightarrow 4k > -20 \Rightarrow k > -5$$

5. 圓: $x^2 + y^2 - 6x + 8y + 0 = 0$

$$\text{令 } d = -6, e = 8, f = 0$$

$$\text{圓心 } \left(-\frac{d}{2}, -\frac{e}{2}\right) = \left(-\frac{-6}{2}, -\frac{8}{2}\right) = (3, -4)$$

$$\text{半徑} = \frac{1}{2} \sqrt{d^2 + e^2 - 4f}$$

$$= \frac{1}{2} \sqrt{(-6)^2 + 8^2 - 4 \times 0}$$

$$= \frac{1}{2} \sqrt{100} = \frac{1}{2} \times 10 = 5$$

$$\text{故圓的參數式為 } \begin{cases} x = 3 + 5\cos\theta \\ y = -4 + 5\sin\theta \end{cases}, 0 \leq \theta < 2\pi$$

6. $\because$ 點 $(4, 3)$ 代入圓 $C: 4^2 + 3^2 - 8 \times 4 + 7 = 0$

$\therefore$ 點 $(4, 3)$ 在圓上

7. 圓 C 的圓心 $Q\left(-\frac{2}{2}, -\frac{-4}{2}\right) = (-1, 2)$

$$\text{半徑 } r = \frac{1}{2} \sqrt{2^2 + (-4)^2 - 4 \times (-4)} = 3$$

圓心 Q 到直線 L 的距離

$$b = \frac{|3 \times (-1) + 4 \times 2 + 5|}{\sqrt{3^2 + 4^2}} = \frac{10}{5} = 2 < r$$

則直線 L 與圓 C 有 2 個交點，即 $a = 2$

因此 $a + b = 2 + 2 = 4$

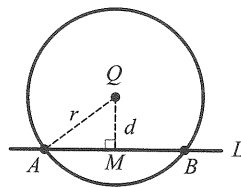
8. 圓的圓心 $Q\left(-\frac{2}{2}, -\frac{-6}{2}\right) = (-1, 3)$

$$\text{半徑 } r = \frac{1}{2} \sqrt{2^2 + (-6)^2 - 4 \times (-5)} = \sqrt{15}$$

圓心 Q 到直線 $L: 3x + 4y + 1 = 0$ 的距離

$$d = \frac{|3 \times (-1) + 4 \times 3 + 1|}{\sqrt{3^2 + 4^2}} = 2$$

如圖



在 $\triangle AQM$ 之中

$$r^2 = d^2 + \overline{AM}^2 \Rightarrow (\sqrt{15})^2 = 2^2 + \overline{AM}^2$$

$$\Rightarrow \overline{AM} = \sqrt{11}, \text{ 因此 } \overline{AB} = 2\overline{AM} = 2\sqrt{11}$$

9. 圓: $(x+2)(x+2) + (y-5)(y-5) = 10$

把點 $(1, 6)$ 代入上式有底線的 x, y

就是 L 的方程式

$$(1+2)(x+2) + (6-5)(y-5) = 10$$

$$\Rightarrow 3(x+2) + (y-5) = 10$$

$$\Rightarrow 3x + y - 9 = 0$$

10. 圖： $x \times x + y \times y + (x+x) - (2y+2y) - 20 = 0$

把點(3,5)代入上式有底線的 x 、 y

就是切線方程式

$$3 \times x + 5 \times y + (3+x) - (2 \times 5 + 2y) - 20 = 0$$

$$\Rightarrow 3x + 5y + 3 + x - 10 - 2y - 20 = 0$$

$$\Rightarrow 4x + 3y - 27 = 0$$

12 單元6 數列與級數

(主題 1~9：等差數列與等差級數)

答案

- | | |
|--------|----------|
| 1. 17 | 2. 128 |
| 3. 705 | 4. 97 |
| 5. 35 | 6. 18 |
| 7. 504 | 8. 255 |
| 9. 140 | 10. 1335 |

$$\begin{aligned} 1. \quad & \sum_{k=1}^5 (2^k - 9) \\ &= (2^1 - 9) + (2^2 - 9) + (2^3 - 9) + (2^4 - 9) + (2^5 - 9) \\ &= (2 - 9) + (4 - 9) + (8 - 9) + (16 - 9) + (32 - 9) \\ &= -7 + (-5) + (-1) + 7 + 23 = 17 \end{aligned}$$

$$\begin{aligned} 2. \quad & \sum_{k=1}^{100} \left(\frac{a_k}{5} - \frac{b_k}{2} + 1 \right) = \frac{1}{5} \sum_{k=1}^{100} a_k - \frac{1}{2} \sum_{k=1}^{100} b_k + \sum_{k=1}^{100} 1 \\ &= \frac{1}{5} \times 205 - \frac{1}{2} \times 26 + 100 \times 1 = 128 \end{aligned}$$

3. 設此等差數列的第 n 項為 a_n
則首項 $a_1 = 5$ ，公差 $d = 12 - 5 = 7$
而 $a_{101} = a_1 + (101 - 1)d = 5 + 100 \times 7 = 705$

4. 設此等差數列的公差為 d
 $\therefore a_{15} = a_9 + (15 - 9)d$
 $\therefore 100 = 58 + 6d \Rightarrow d = 7$
又設 674 是此等差數列的第 k 項 a_k
則 $a_k = a_{15} + (k - 15)d$
 $\Rightarrow 674 = 100 + (k - 15) \times 7$

$$\begin{aligned} &\Rightarrow 7(k - 15) = 574 \xrightarrow{+7} k - 15 = 82 \\ &\Rightarrow k = 97 \end{aligned}$$

5. 設等差數列 $\langle a_n \rangle$ 的 $a_1 = -5$ ， $a_7 = 19$

公差為 d

$$\text{則 } a_7 = a_1 + (7 - 1) \times d$$

$$\Rightarrow 19 = -5 + 6d \Rightarrow d = 4$$

$$\text{故 } a_2 = -5 + 4 = -1, a_3 = -1 + 4 = 3, a_4 = 3 + 4 = 7,$$

$$a_5 = 7 + 4 = 11, a_6 = 11 + 4 = 15$$

$$\text{即： } m = -1, n = 3, p = 7, q = 11, r = 15$$

$$\text{因此 } m + n + p + q + r = -1 + 3 + 7 + 11 + 15 = 35$$

6. (1) $\therefore 3a - 4, 26, 5a$ 是等差數列

$$\therefore 26 = \frac{(3a - 4) + 5a}{2}$$

$$\Rightarrow 26 = 4a - 2 \Rightarrow 4a = 28$$

$$\Rightarrow a = 7$$

(2) $\therefore a$ 與 15 的等差中項是 b

$$\therefore b = \frac{a + 15}{2} = \frac{7 + 15}{2} = \frac{22}{2} = 11$$

$$\text{由(1)與(2)，則 } a + b = 7 + 11 = 18$$

7. 設等差數列 $\langle a_n \rangle$ 的 $a_1 = 20$ ， $a_2 = 24$

公差 $d = 4$

$$\begin{aligned} \text{數列的前 12 項和} &= \frac{12 \times [2a_1 + (12 - 1)d]}{2} \\ &= \frac{12 \times (2 \times 20 + 11 \times 4)}{2} = 504 \end{aligned}$$

故集會場共有 504 個座位

8. 設此等差級數的第 n 項為 a_n ，公差為 d

$$\text{則 } a_1 = 3, d = 5 - 3 = 2$$

$$\text{若第 } n \text{ 項 } a_n = 31, \text{ 則 } a_n = a_1 + (n - 1)d$$

$$\Rightarrow 31 = 3 + (n - 1) \times 2 \Rightarrow n = 15$$

故此等差級數共有 15 項

$$\text{其和} = \frac{15 \times (3 + 31)}{2} = \frac{15 \times 34}{2} = 255$$

$$\begin{aligned} 9. \quad & a_1 + a_2 + a_3 + \cdots + a_{20} \\ &= \frac{2 \times 1}{3} + \frac{2 \times 2}{3} + \frac{2 \times 3}{3} + \cdots + \frac{2 \times 20}{3} \\ &= \frac{2}{3} \times (1 + 2 + 3 + \cdots + 20) \\ &= \frac{2}{3} \times \frac{20 \times (20 + 1)}{2} = 140 \end{aligned}$$

$$\begin{aligned} 10. \quad & \sum_{k=1}^{30} (3k - 2) = 3 \sum_{k=1}^{30} k - \sum_{k=1}^{30} 2 \\ &= 3 \times \frac{30 \times (30 + 1)}{2} - 30 \times 2 \\ &= 1395 - 60 = 1335 \end{aligned}$$